ECO4108Z CREDIT DERIVATIVES PRACTICE QUESTIONS SOLUTIONS

Question 1

a. i. Following several changes in BK Ltd's liquidity, debt-to-equity ratio and interest coverage, Moody's updates their credit rating from Baa to Aa. Comment on this new credit rating and on how you would anticipate investors to react.

(2 marks)

[redacted]

ii. What type of credit risk does the above scenario relate to?

(1 mark)

[redacted]

iii. A year from now you notice that BK Ltd has a rating of "Withdrawn", comment on what this rating could mean?

(1 mark)

[redacted]

Question 2

a. Corvus Bank has several clients in the Hospitality and Tourism industry that they have issued loans to. The total value of these assets on their balance sheet amounts to R30 million. Corvus Bank is concerned that these clients will struggle to maintain consistent revenue streams needed to repay their loans, given the volatile fluctuations observed in consumer spending and changes in travel patterns. If Corvus Bank is also in need of a capital infusion and they are willing to transfer ownership of their assets, which type of CDO would you recommend to them?

(2 marks)

[redacted]

b. Based on your recommendation in (a) above, calculate the amount of regulatory capital that has been freed up? Assume that banks are required to maintain risk-based capital of 11,5% of the outstanding balance of commercial loans (R30 million in this case). Assume further that Corvus Bank must also retain a 5% subordinate position in the CDO type you recommended above, where regulation requires that R1,20 in regulatory capital be held for every R1 of ownership in the CDO.

(4 marks)

[redacted]

c. Corvus Bank also issues a commercial loan to a startup, Tatooine Technologies, for an amount of R25 million. Startups are often viewed as high-risk borrowers because they are unproven and may not have a track record of generating revenue. Corvus Bank decides to take out a Credit Default Swap with Rainy Day Asset Management to buy credit protection in the event that Tatooine Technologies may not be able to meet its debt obligations as they come due.

i. Based on the three groupings of credit derivatives that we discussed in class, how would you classify this Credit Default Swap?

(3 marks)

[redacted]

ii. Suppose that the CDS has a tenor of 4 years, a CDS spread of 6% per year, payable quarterly, and that settlement in the event of a trigger event would be physical, i.e. R25 million in face value issued in return for a R25 million payment. If Tatooine Traders fails to make payment on their debt after two and a half years, what cash flows and exchanges would take place?

(4 marks)

[redacted]

Question 3

a. Paz Vizsla is a CDO manager with a collateral portfolio of high yield bonds valued at R85 million and an average yield of 9%. He wants to earn a positive spread between the interest paid to CDO investors and the return from the CDO assets. One of the tranches in this CDO is rated BB has attachment and detachment points respectively, of 5%/33% and has a yield of 7%. Another tranche has a rating of AAA, a yield of 4% and a tranche width of 67%. Paz charges an annual management fee of 65 basis points for managing the collateral portfolio and also has a 5% equity stake in the CDO. Calculate the return of the junior tranche, which is unrated and collects any residual income generated by the CDO collateral, assuming no defaults.

(10 marks)

[redacted]

b. If the CDO assets experience 40% default with 12% recovery, what would happen to the collateral portfolio and each of the three tranches?

(5 marks)

[redacted]

c. Comment on whether the Senior Tranche investor in this CDO would prefer if the reference portfolio's assets had a diversity score of 5 or 75? Motivate your answer.

(2 marks)

[redacted]

d. Define overcollateralization in the context of credit risk enhancements and assess the overcollateralization of each of the tranches in this CDO.

(4 marks)

[redacted]

Question 4

In the context of regulation in credit derivatives, do you agree or disagree with the critique that instead of reducing systemic risk in the banking sector, credit risk transfer contributes to an increase in systemic risk in the banking sector? Provide an explanation for your answer.

(3 marks)

[redacted]

Question 5

Suppose your friend is viewing the equity tranche of a Collateralised Debt Obligation's (CDO) reference portfolio as a call option and is interested in the default interconnectedness among the assets held in the reference portfolio. In your own words, how would you explain the concept of Diversity scores to them?

(3 marks)

[redacted]

[redacted]

Question 6

An investor wants to hedge her credit risk. The referenced entity is a biotech firm and the referenced asset has R25 million of face value debt. She takes out a CDS agreement with a bank with a maturity of 6 years. In exchange for protection, the investor pays 4.5% of the notional amount per year, payable quarterly. The contract will be settled physically. If default occurs after exactly three years, what cash flows and exchanges would take place?

(3 marks)

[redacted]

Question 7

Suppose a CDO is being used to structure cash flows from a portfolio of high yield bonds. There are R40 000 000 of high yield bonds serving as the collateral portfolio for the structure with an average yield of 11%. One of the tranches in the structure has a BBB rating and a yield of 8% and contributes 33% of the financing. Another tranche is rated AA, has a yield of 5% and represents 55% of the structure. The balance of the structure's financing is from the subordinated tranche which is not rated and is the residual claimants of the CDO structure.

a. If the CDO manager charges an annual management fee of R250 000 and there are no defaults, calculate how much annual income should be distributed to the subordinated tranche. Show all your workings.

(10 marks)

[redacted]

b. Suppose that the CDO experiences defaults in R12 000 000 of the assets with 30% recovery. Explain what would happen to the collateral portfolio, to the annual income and to each of the tranches.

(5 marks)

[redacted]